

Logistic Regression

Overview

- a **machine learning** task where we **predict** a real-valued **output** response given an **input vector**
- **linear** and **non-linear** regression **models**
- **analytic** (closed-form) **solution** to learn an **optimal model** from **training data**
- **gradient descent** and **stochastic gradient** descent algorithms
- model **complexity** and **overfitting**

Readings

- <https://www.microsoft.com/en-us/research/wp-content/uploads/2006/01/Bishop-Pattern-Recognition-and-Machine-Learning-2006.pdf#page=157>

Async	
Logistic Regression	
Definition	• supervised learning technique used to train models for classification where y is categorical • generally, links to a binary link; is $x[a, b, c, \dots]$ likely to produce y • $x \in \mathbb{R}^d \Rightarrow y \in \{1, \dots, C\}$
Fitting the Model	• maximize the likelihood of y given x • $P_{\theta}(y = 1 x) = \sigma(\varphi(x)^T \theta) = \frac{1}{(1 + \exp(-\varphi(x)^T \theta))}$ • to maximize the likelihood by minimizing the negative log likelihood (NLL) • $\operatorname{argmax}_{\theta} \prod_{i=1}^n P_{\theta}(y_i x_i) =$ $\operatorname{argmin}_{\theta} - \frac{1}{n} \sum_{i=1}^n \log P_{\theta}(y_i x_i)$ • putting the likelihood of y given x into terms of the regression model ○ for $y = 1$, we need the probability of $y = 1$ given x given by the sigmoid of the linear function: $\sigma(\theta^T x_i)$ ○ for $y = 0$, we need the probability of $y = 0$ given x is given by $1 - \sigma(\theta^T x_i)$ ○ which implies an 'if' statement in our algorithm ○ so, $\log p_{\theta}(y_i x_i) = \begin{cases} \log \sigma(\theta^T x_i) & \text{if } y_i = 1 \\ \log(1 - \sigma(\theta^T x_i)) & \text{if } y_i = 0 \end{cases}$ ○ utilizing indicator values as coefficients, we can effectively insert a Boolean 'if' in the equation ○ so, $\log p_{\theta}(y_i x_i) =$ $\{y_i = 1\} \log \sigma(\theta^T x_i) +$

$$\mathbb{1}\{y_i = 0\} \log(1 - \sigma(\theta^T x_i))$$

$$\text{given } \begin{cases} \mathbb{1} = 1 & \text{if True} \\ \mathbb{1} = 0 & \text{if False} \end{cases}$$

- and finally, “after some algebra”, we conclude that

$$\arg \min_{\theta} - \frac{1}{n} \sum_{i=1}^n (y_i \varphi(x_i)^T \theta + \log(\sigma(-\varphi(x_i)^T \theta)))$$

is equivalent to minimizing a KL divergence known as the cross-

entropy loss

- this justifies the **binary adjustment** to the **classification problem**